Chapter 1

Introduction

These notes serve as an introduction to public finance and budgeting. At the end of each chapter, there are exercises to practice the concepts covered. The final chapter provides case studies that require more in-depth analysis than the exercises. The chapters are subdivided into five parts of varying length: (1) Introduction and role of government in the economy, (2) budgeting, (3) cost analysis and capital budgets, (4) government revenue, and (5) macroeconomic perspective of government finances.

Introduction and role of government in the economy: This part serves as an introduction to government finance and budgets. The topics covered in this part are federal government, state/local government, efficiency, and real versus nominal prices as well as the role of government in the economy. A major emphasis is placed on the purpose of taxation and government intervention in the economy. Topics such as externalities, public goods, and imperfect competition (commonly referred to as market failures) are discussed, all of which justify (to a certain degree) government's role in the economy. The chapter also covers economic stabilization and redistribution as reasons for government intervention.

Budgeting: This part starts with recurrent budgets that cover current-year expenditures with current-year revenue. A recurrent budget is different from capital budgeting, which is covered later in the notes. A recurrent budget provides a framework for matching recurring revenues with recurring expenditures and for assessing the government's short-term fiscal position. We cover the budgeting process, which typically involves a cycle that includes budget preparation, review, approval, and execution. This ensures that each phase is planned and scrutinized before implementation, allowing for effective management of public funds. Since these notes focus on budgeting in the United States, several specific topics related to federal, state, and local budgets are covered. Institutions, audits, and evaluations are critical for ensuring transparency and accountability throughout the process. Appropriation bills are required for discretionary spending, which funds specific programs and services. Meanwhile, mandatory spending, such as Social Security and Medicare, does not require appropriation bills, as it is governed by existing laws. The last aspects covered in this part are budget methods and formats, which vary depending on the needs of the government or institution. Budgets are often classified by categories such as functions, programs, or departments. The budget format typically includes examples and breakdowns to show how funds are allocated across different areas, ensuring clarity in the decision-making process.

Cost analysis and capital budgeting: Cost analysis is essential to ensure that resources are used efficiently and that decisions are based on a thorough understanding of the financial impact. Institutions need to know the cost associated with service provision to prepare a budget. This chapter covers cost accounting and estimation as well as cost allocation. Purchasing assets that are used for multiple years (e.g., bridges, public school buildings, infrastructure in general) require assessment of future costs and benefits, which are covered in this chapter. The characteristics of capital assets are that (1) their useful life is more than one year, (2) the expenditure is non-recurring, (3) the expenditure is usually very large (e.g., infrastructure projects such as bridges or airports).

Government revenue: Compared to the previous chapters, which focus on the expenditure side of government activities, this part introduces the revenue side. General aspects of taxation such as equity, adequacy of revenue generation, the ability to collect revenue, and economic effects are covered. The various revenue-generating activities presented are income taxes (i.e., taxes on individual and corporate incomes), taxes on goods and services (e.g., sales tax, fuel tax), property taxes, and other revenue sources (e.g., license fees, lotteries). Property taxes represent the majority of revenue for local governments and are spent locally on education or public safety. At the state and local level, governments rely on three main sources of revenue: taxes, federal aid, and borrowing. These sources enable them to fund services and infrastructure projects.

Macroeconomic perspective of government finances: The first sections covers debt management and forecasting. At the federal level, it is important to distinguish between a deficit or surplus and debt. A deficit occurs when the government's outlays in a given fiscal year exceed its revenues, while a surplus arises when revenues exceed expenditures. In contrast, debt refers to the total amount of money the government owes, accumulated over years of deficits. A key aspect of borrowing involves bonds. When a government or entity issues a bond, it commits to paying the bondholder regular interest and repaying the principal at a future date, known as the maturity date. There are various types of bonds, and their values fluctuate based on interest rates, market demand, and the creditworthiness of the issuer. Deficit and debt management involve strategies to handle both short-term fiscal imbalances and long-term accumulated debt. This ensures that governments can meet their financial obligations while maintaining economic stability.

1.1 Public Finance and Artificial Intelligence

Although AI will likely fundamentally change the work conducted by public managers in the future, understanding public finance and budgeting remains essential because the issues addressed are ultimately about values and institutions and not only calculation. AI can forecast revenue, model the incidence of a tax, or optimize the allocation of a fixed budget across competing programs, but it cannot evaluate by itself how much a society should redistribute income, which public goods deserve priority, or how much risk future generations should bear for present spending. These are normative and political judgments that citizens and their elected representatives make. Public finance is ultimately about the collective choices a society makes regarding what to fund, whom to tax, and how to manage risk across time, and no algorithm can substitute for the informed judgment that democratic governance of those choices requires. As AI lowers the cost of producing fiscal analysis or putting together a budget, the ability to ask the right questions of a model's output, to recognize its assumptions, and to verify its conclusions against basic economic principles becomes more important.

1.2 Data Sources

The intention of these notes is to work with data sources that are updated on a regular basis. This allows the reader to obtain updated data for the information and concepts provided. To the extent possible, all data is presented from the original source. For example, you can ask AI about the current rate of inflation but the actual data, which AI or any other outlet relies on, is provided by the Bureau of Labor Statistics (BLS). The following presents some of the main data sources used in the notes.

- **Federal Reserve Economic Data (FRED):** A data source extensively used throughout these notes is the Federal Reserve Economic Data (FRED) of the St. Louis Fed. The abovementioned data series regarding the debt held by the public as a percentage of GDP is obtained from FRED.
- **Office of Management and Budget (OMB):** Many data series associated with the federal budget are obtained from the White House's OMB. You can search for the historical tables and find not only all the historical data but also the projections for the next couple of years.
- **Congressional Budget Office (CBO):** The CBO is associated with the legislative branch and publishes reports on a plethora of budgetary topics such as the CBO Long Term Budget Analysis or the cost of the Coast Guard's Polar Security Cutter. Of course, the long-term budget analysis is of great interest to stakeholders since it quantifies federal revenues and expenditures given current policies and macroeconomic projections. Those projections may not always be accurate but can serve a guidance for policy making. For example, in the 2002 CBO Projections (January 2002) The Budget and Economic Outlook: Fiscal Years 2003-2012, the projected debt held by the public as a percentage of GDP in 2012 was

estimated to be 7.4% but the actual debt was 69.4% (Source). Significant discrepancy between projection and reality can occur due to policy changes and unforeseen shocks such as tax cuts, homeland security, 2001 economic downturn, 2008–2009 Great Recession, stimulus spending (all in the 2000s), or COVID-19.

There are many other data sources out there. For more micro-level data, the U.S. Census Bureau is a very good source. International organizations such as the International Monetary Fund (IMF), the World Bank, or the Organisation for Economic Co-operation and Development (OECD) provide data for all countries on important economic indicators. For example, data on general government gross debt as a percentage of GDP is provided by the IMF. The OECD compares the tax revenue as a percent of GDP among member nations. For the European Union (EU), the main data source is EUROSTAT. For the EU members as well as some member states, you can obtain, for example, the general government deficit/surplus or the general government gross debt.

1.3 Size of the Federal Government

This chapter serves as an introduction to budgeting and public finance. We quantify the size of the U.S. government relative to its economy as well as give an international perspective. We show receipt and expenditure trends in the U.S. by bearing in mind that those numbers change every year and can be impacted by major events such as the COVID-19 pandemic. A term that is used often throughout the book is Gross Domestic Product (GDP). The Bureau of Economic Analysis (BEA) defines GDP as follows:

> *The value of the goods and services produced in the United States is the gross domestic product.*

Economic growth is measured by changes in the GDP and is an important macroeconomic indicator. GDP growth (or lack thereof) serves as an important indicator for recession dating besides employment levels. The National Bureau of Economic Research (NBER) announced the following by its Business Cycle Dating Committee on January 7, 2008:

> *We view real GDP as the single best measure of aggregate economic activity. In determining whether a recession has occurred and in identifying the approximate dates of the peak and the trough, we therefore place considerable weight on the estimates of real GDP issued by the Bureau of Economic Analysis (BEA) of the U.S. Department of Commerce.*

Let us next present some broad numbers illustrating the size of the U.S. government in terms of expenditures (also called outlays) and revenue. Those numbers are taken from the CBO's Federal Budget 2025. There are two broad categories of outlays. Discretionary outlays (e.g., defense) go through the annual appropri-

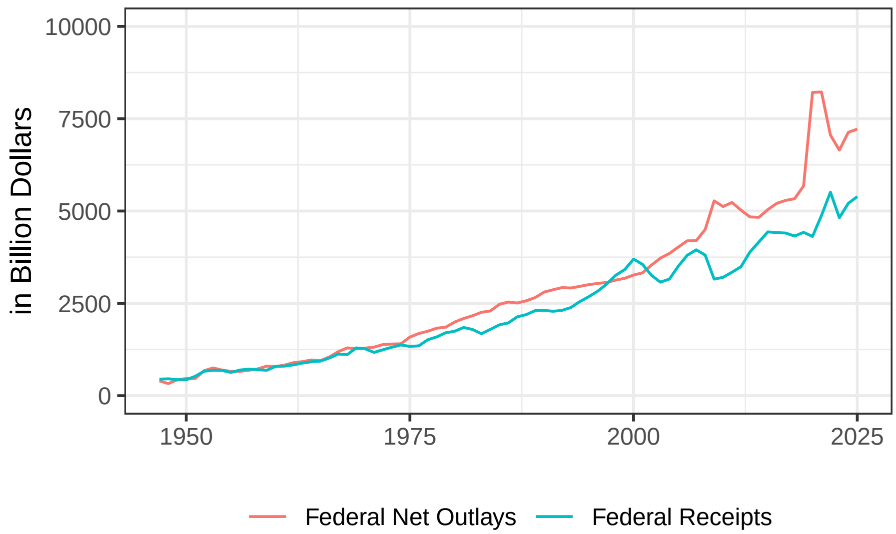


Figure 1.1: Real federal outlays and receipts in constant 2017 USD. Source: FRED Federal Reserve Economic Data St. Louis Fed.

ation process. As opposed to the discretionary outlays, mandatory outlays (e.g., Social Security, Medicare, Medicaid) do not go through the appropriation process. The total government revenue in 2025 was $5.2 trillion from the following, broad sources:

- Individual income taxes ($2.7 trillion)
- Payroll taxes such as Social Security ($1.7 trillion)
- Corporate income taxes ($452 billion)
- Other revenues such as excise taxes, estate and gift taxes, customs duties

Government expenditures (outlays) amounted to $7.0 trillion in the following categories:

- Mandatory spending ($4.2 trillion)
- Discretionary spending ($1.9 trillion)
- Net interest ($970 billion)

Note the difference (deficit) between the expenditures and revenue of about $1.8 trillion. Below are the federal net outlays and receipts in real terms, that is, adjusted for inflation.

Although the previous graph suggested a significant increase in outlays and revenue, the size of the U.S. government as a percentage of GDP has been relatively stable over time given the Federal Receipts as Percent of Gross Domestic Product, Federal Net Outlays as Percent of Gross Domestic Product, and Federal

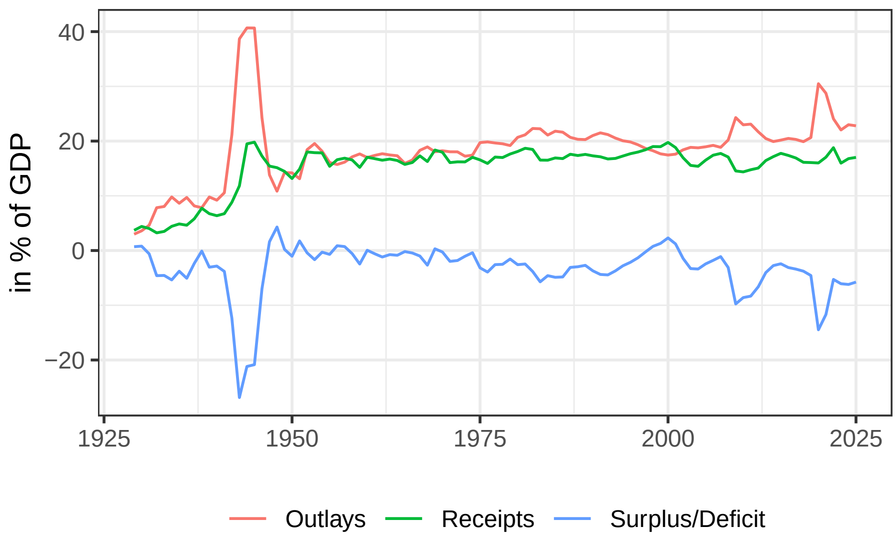


Figure 1.2: Outlays, receipts, and surplus/deficit as a percentage of GDP.

Surplus or Deficit as Percent of Gross Domestic Product.

Given the deficit and associated debt, which needs to be financed, there is often a discussion regarding which countries hold how much of the U.S. debt. For January 2023, the Major Foreign Holders of Treasury Securities (in billion USD) were the following countries:

Top 10 Countries	Jan-23	GDP	Ratio
Japan	$1,104	$4,231	26.1%
China	$859	$17,963	4.8%
UK	$668	$3,071	21.8%
Belgium	$331	$579	57.2%
Luxembourg	$318	$82	387.8%
Switzerland	$291	$ 808	36.0%
Cayman Islands	$285		
Canada	$254	$2,140	11.9%
Ireland	$253	$529	47.8%
Taiwan	$235		

1.4 State and Local Governments

Besides the federal government, there is a large number of local governments in the United States. According to the U.S. Census Bureau 2017 Census of Governments

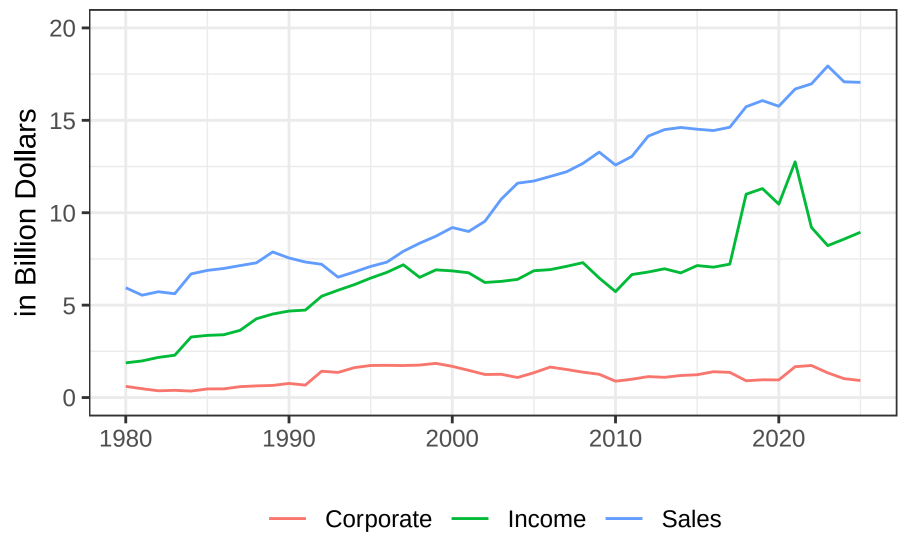


Figure 1.3: Real state government tax collections for the state of Indiana. Source: U.S. Bureau of Census Annual Survey of State Government Tax Collections (STC).

Organization: 50 State governments, 19,495 municipal governments, 16,253 town or township governments, 3,031 county governments, 12,754 independent school districts, and 38,542 special district governments. Those numbers are relatively stable over time. Some taxes only exist at certain levels of government while others can exist at multiple levels. For example, there are fuel taxes at the federal and state level and sometimes even at the local level. Below is an example of the real tax revenue for Indiana.

1.5 Evaluating Government Decisions: Efficiency

Government activity either at the federal, state, or local level should be evaluated in terms of social benefits and costs. The term "efficiency" is often used in economics and public finance. In economics, it refers to the allocation of resources such that overall welfare to society is maximized. In the context of public finance, let us focus for now on Pareto or Kaldor-Hicks efficiency. Pareto efficiency is achieved when no person can be made better off without another person being made worse off. Since this is very restrictive, the Kaldor-Hicks efficiency offers a less stringent alternative. Under Kaldor-Hicks improvement, there is a positive net benefit to society such that winners could (at least theoretical) compensate the losers after a policy change. Although Kaldor-Hick efficiency is usually the basis for a cost-benefit analysis, the drawback is that the compensation is potential and may never be realized. Consider the table with the individual benefits and costs associated with a single project.

Individual Resident	Individual Benefit	Cost Share	Individual Gain
A	\$3,500	\$2,000	\$1,500
B	\$1,000	\$2,000	(\$1,000)
C	\$1,500	\$2,000	(\$500)
D	\$4,500	\$2,000	\$2,500
E	\$1,500	\$2,000	(\$500)
Total Benefit	\$12,000	\$10,000	\$2,000

The project is a Kaldor-Hicks improvement because the total benefit (\$12,000) exceeds the total cost (\$10,000). However, the project is not Pareto efficient because the move from the status quo to the project is not a Pareto improvement. This project would also not pass a majority vote because A and D gain, while B, C, and E lose, so the vote would be 2 versus 3.

1.6 Comparing Dollar Values Across Time: Nominal and Real Values

Evaluating government budgets over time requires adjusting dollar amounts for inflation. Historical prices can appear remarkably low when viewed in isolation, but nominal (i.e., expressed in the dollar value of the time) prices by themselves tell us little about how expensive goods actually were. A poster outside a convenience store in Graeagle (California) listed prices in 1918. For example, it reports that a loaf of bread cost \$0.10, a gallon of gasoline \$0.15, a gallon of milk \$0.56, a new car \$440, and a new house \$4,821. At first glance, these prices seem extraordinarily cheap by today's standards. However, the same poster reports an average annual income of only \$1,697. The relevant economic question is therefore not simply how many dollars a good cost, but how large that cost is relative to income and the general price level at the time. A \$440 automobile in 1918 represented more than one-quarter of average annual income, illustrating why directly comparing nominal dollar amounts across widely separated years can be misleading. Inflation changes the purchasing power of money, so a dollar in 1918 is not economically equivalent to a dollar today. To make meaningful comparisons across time, economists therefore convert nominal prices and incomes into real values, which hold the purchasing power of money constant and allow us to distinguish changes in affordability and economic well-being from changes that merely reflect inflation.

One way to understand the distinction between nominal and real values is to compare income and prices across time. Suppose that in period 1, income is \$100, apples cost \$5 each and four apples are purchased, while milk costs \$10 per gallon and eight gallons are purchased. In period 2, income rises to $120, but the prices of apples and milk also rose to $6 and $12, respectively. The quantities purchased remain unchanged. Although nominal income has increased by 20%, the individual's purchasing power has not increased because prices have also risen by 20%. In this sense, the $120 income in period 2 has the same real value as the $100 income in period 1. An income of $110 in period 2 would therefore represent a decline in real purchasing power, even though it is nominally higher than the $100 received in period 1.

Economists use index numbers to make such comparisons systematically. An index number combines information from several individual values into a single measure, while an index is a series of such numbers used to track changes over time. Common examples include stock market indices such as the Dow Jones Industrial Average and the S&P 500, as well as the Consumer Price Index (CPI).

The CPI, published by the Bureau of Labor Statistics (BLS), measures the average change over time in the prices paid by urban consumers for a representative market basket of goods and services. The BLS defines the CPI as follows:

> *The Consumer Price Index (CPI) is a measure of the average change over time in the prices paid by urban consumers for a market basket of consumer goods and services.*

This basket contains thousands of items grouped into categories such as housing, food and beverages, transportation, medical care, apparel, recreation, education and communication, and other goods and services. The CPI in period (t) is calculated as follows:

$$CPI_t = \frac{MB_t}{MB_b} \cdot 100$$

where MB_t is the cost of the market basket in the period of interest and MB_b is its cost in the base period. For example, if the basket costs $71 in the current period and $68 in the base period, the CPI is approximately 104.41, indicating that the general price level is about 4.41% higher than in the base period. The CPI is therefore useful as an economic indicator, as a measure for adjusting payments such as pensions, and especially as a deflator for converting nominal dollar values into real values that reflect purchasing power. The percentage change of the CPI over the past 12 months is commonly used to measure inflation. The inflation rate can be expressed as follows with t being measured in months:

$$\pi_t = \frac{CPI_t - CPI_{t-12}}{CPI_{t-12}} \cdot 100$$

When monthly CPI data are reported in the news, inflation is often presented as the percentage change relative to the same month one year earlier. For example, the CPI in July 2025 and July 2026 measures 323.048 and 333.918, respectively. The increase of 3.365% represents the the 12-month inflation rate for July 2026. Although widely used, the CPI is not a perfect measure of the cost of living because it can be affected by substitution bias, the introduction of new technologies and products, changes in product quality, and consumers shifting toward lower-cost or discount retailers. Ultimately, the distinction between nominal and real values is important because nominal dollar amounts alone do not indicate economic well-being; what matters is the quantity of goods and services those dollars can purchase.

1.6.1 Headline Inflation vs. Core Inflation

Inflation can be measured in different ways depending on which goods and services are included in the price index. Headline inflation measures the change in the overall price level and therefore includes all major components of consumer spending, including food and energy. These two categories, however, tend to be especially volatile. Food prices can change sharply because of events such as droughts, floods, or other weather shocks that reduce crop yields, while energy prices can fluctuate substantially because of changes in global oil supply, including production decisions by OPEC. For this reason, economists also examine core inflation, which excludes food and energy prices in order to provide a measure of underlying inflationary pressure that is less affected by short-term price swings. For example, the CPI increased from 294.628 in July 2022 to 304.348 in July 2023, corresponding to a headline inflation rate of approximately 3.3 percent. This figure reflects the overall increase in consumer prices, including changes in food and energy prices, whereas the corresponding core inflation measure would exclude those two categories. Comparing headline and core inflation can therefore help distinguish temporary price movements from more persistent changes in the general price level.

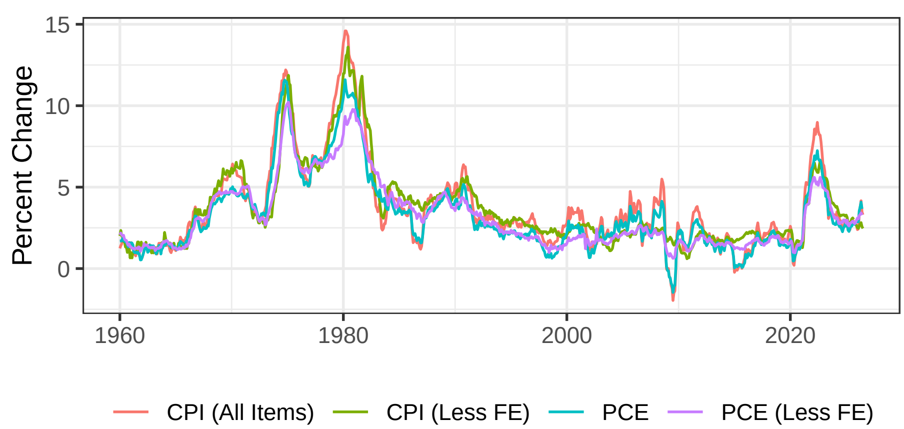
Two other important measures of the overall price level are the Personal Consumption Expenditures (PCE) Price Index and the GDP deflator. The PCE Price Index measures changes in the prices of goods and services consumed by households and is produced by the Bureau of Economic Analysis. One major difference from CPI is that PCE includes expenditures made on behalf of households, such as employer- or government-financed health care. It differs from the Consumer Price Index in several ways, including broader coverage of consumer expenditures and a weighting system that adjusts more readily when consumers substitute away from goods whose relative prices rise. For this reason, the PCE Price Index is often viewed as a broader measure of consumer inflation and is the inflation measure emphasized by the Federal Reserve when evaluating progress toward its price-stability objective. A footnote in the Monetary Policy Report to the Congress (17 February 2000) states the following:

> *The chain-type price index for PCE draws extensively on data from the consumer price index but, while not entirely free of measurement problems, has several advantages relative to the CPI. The PCE chain-type index is constructed from a formula that reflects the changing composition of spending and thereby avoids [. . .] fixed-weight nature of the CPI. In addition, the weights are based on a more comprehen-*

> *sive measure of expenditures. Finally, historical data used in the PCE price index can be revised to account for newly available information and for improvements in measurement techniques, including those that affect source data from the CPI; the result is a more consistent series over time.*

The GDP deflator, in contrast, measures the prices of all domestically produced final goods and services included in gross domestic product. Because it covers consumption, investment, government purchases, and exports while excluding imports, it is broader than either the CPI or the PCE Price Index. The GDP deflator is calculated as the ratio of nominal GDP to real GDP multiplied by 100 and is particularly useful for distinguishing changes in the dollar value of total economic output that result from changes in production from those that result simply from changes in prices.

The CPI, PCE Price Index, and GDP deflator all measure changes in prices, but they serve somewhat different purposes. The CPI focuses on prices paid directly by consumers and is commonly used to measure changes in the cost of living and to adjust wages, benefits, and other payments for inflation. The PCE Price Index also measures consumer prices but covers a broader range of expenditures and adjusts more readily for changes in consumer purchasing patterns; it is the inflation measure emphasized by the Federal Reserve. The GDP deflator is broader still, measuring price changes for all domestically produced final goods and services, and is mainly used to convert nominal GDP into real GDP and assess economy-wide price changes.



Because nominal dollar values combine changes in prices with changes in quantities, economists often convert them into constant dollars, or real values, before making comparisons across time. A nominal value observed in period (t) can be expressed in the purchasing power of a base period (b) using

$$CD_b = \frac{ND_t \cdot CPI_b}{CPI_t}$$

For example, gasoline that cost \$0.84 per gallon in May 1979 may appear extremely inexpensive compared with current prices, but after adjusting for inflation using a CPI of 71.4 in May 1979 and 288.663 in April 2022, that price is equivalent to about \$3.40 in 2022 dollars. This illustrates why nominal comparisons can be misleading: part of any change in a dollar amount may simply reflect inflation rather than a real change in economic activity. More generally, the growth of a nominal economic series can be decomposed into a real component, reflecting changes in the quantity of goods and services, and a price component, reflecting inflation. For instance, federal national defense consumption expenditures and gross investment increased from \$827.971 billion in 2010 to \$847.830 billion in 2019 and \$882.443 billion in 2020, but the GDP implicit price deflator also rose from 95.023 in 2010 to 112.315 in 2019 and 113.769 in 2020. Note that we are using the CPI for consumer prices but the GDP deflator for a broader category like defense spending. Therefore, evaluating the true change in defense activity requires adjusting the nominal expenditures for the change in the overall price level rather than interpreting the increase in nominal spending as entirely real growth.

Changes in nominal spending can be separated into changes caused by inflation and changes in the actual quantity of goods and services purchased. Between 2019 and 2020, the GDP implicit price deflator increased from 112.315 to 113.769, implying an inflation rate of approximately 1.29%, while nominal federal defense consumption expenditures and gross investment increased from \$847.830 billion to \$882.443 billion, or 4.08%. To determine how much of this increase represented real growth, the 2019 expenditure can be converted into 2020 dollars using the GDP deflator, yielding approximately \$858.806 billion. Comparing this inflation-adjusted value with 2020 spending shows that real defense expenditures increased by about 2.75 percent. More generally, nominal growth (g), inflation (h), and real growth (m) are related by $1 + g = (1 + h) \cdot (1 + m)$, which shows that nominal growth reflects both changes in prices and changes in real economic activity. Over longer periods, growth is often summarized using a compound annual growth rate, $k = (x_1/x_0)^{1/N} - 1$, which expresses the average annual rate that would generate the observed change between an initial and final value. For example, nominal defense spending increased from \$827.971 billion in 2010 to \$882.443 billion in 2020, corresponding to total nominal growth of 6.58% and an average annual nominal growth rate of about 0.64%. However, after accounting for inflation, real spending over the same period actually declined by about 9.92%, again demonstrating why nominal growth alone can provide a misleading picture of changes in economic activity. The price level rose sufficiently over the decade to more than offset the increase in nominal expenditures

1.7 Exercises

1. ***Real vs. Nominal Income*** (*): Assume that your income in year 1 is $200 and that the prices of raspberries and orange juice are $20 and $10, respectively. In year 2, the prices of raspberries and orange juice are $22 and $11, respectively. Assume that you fully exhausted your income in both years to purchase raspberries and orange juice and that you purchase the same quantities of those items in both years. Did your income increase if you earned $210 in year 2? Distinguish between real and nominal income. How much money do you need to earn in year 2 to have the same real income in both years?

2. ***Inflation and Price Indices*** (***): The Bureau of Labor Statistics (BLS) publishes several price indices, including the traditional Consumer Price Index for All Urban Consumers and the Chained CPI. These indices are frequently used in public finance to adjust tax brackets, Social Security benefits, and other government programs for inflation.

    a. What is the difference between the traditional and chained CPI?
    b. Download the two indices from the FRED website (links included above) from January 2005 to the most recent month available as an Excel file. Since both indices have different base years and thus, differ in level, set both of them equal to 100 for January 2005.
    c. Plot the evolution of both indices from January 2005 until the most recent month. What do you observe? If Social Security benefits had been adjusted using the Chained CPI instead of the traditional CPI, what would the percentage difference of benefits be today?
    d. Compute and plot the annual inflation rates implied by each index. How different are the inflation rates over time?

3. ***AI Audit*** (**): Ask an AI tool of your choice the following question: "Federal defense expenditures increased from $847 billion in 2019 to $882 billion in 2020. By what percentage did real defense spending increase?" Report the first response of the AI tool at the beginning of the homework. Note that the purpose of this exercise is not to report an excessive number of AI prompts and answers. Only report the original prompt, the first answer, and, if necessary, the revised prompt and the new AI response as indicated in the questions below:

    a. Explain why the first AI repsonse to the original question is or is not correct.
    b. Use the GDP implicit price deflator to calculate the correct real change in expenditures.
    c. Explain any conceptual errors made by the AI.
    d. If the AI tool made an error, write a better prompt that would make it more likely that an AI system distinguishes between nominal and real growth. Implement that prompt and report the result.

4. ***Core Inflation, Headline Inflation, and Public Perception*** (***): Policymakers often focus on core inflation, which excludes food and energy prices, arguing that it better reflects the underlying trend in inflation. Citizens, however, experience headline inflation, which includes food and energy, more directly in their daily lives. This difference can create tension between technical policy analysis and public perception of fairness.

   a. Download monthly data that forms the basis to calculate headline and core inflation, i.e., CPI for All Urban Consumers and CPI for All Urban Consumers: All Items Less Food and Energy in U.S. City Average from FRED for the period 2010–present.
   b. Compute and plot annual inflation rates for both indices. Highlight periods where headline inflation diverges sharply from core inflation (e.g., during energy price spikes).
   c. Discuss the moral and policy dilemmas. That is, from the perspective of citizens, is it fair to exclude food and energy—essential goods—when adjusting social benefits and/or minimum wage? From the perspective of policymakers, is it reasonable to focus on a more stable measure to ensure long-run fiscal planning? How might this gap between "lived experience" and "technical measurement" influence public trust in government finance? How can this gap be overcome?

5. ***AI-Assisted Policy Analysis*** (***): Suppose Congress is considering whether Social Security benefits should be indexed using headline CPI or core CPI. Ask an AI system to recommend one of the two measures and provide its reasoning. Include the prompt and response. Next, identify the assumptions underlying the recommendation. Ask the AI to make the strongest argument for a different inflation measure. Make your own recommendation and explain why your conclusion agrees or disagrees with the AI.

6. ***Public Good*** (**): A community project (a public good) will cost $2,500 and will benefit the five residents of the community as indicated in the table below. Suppose the local government asks an AI system whether the public project described above should be undertaken.

   a. What information from the table could an AI system use to make its recommendation?
   b. Ask an AI system whether the project should be approved. Record its recommendation and reasoning.
   c. Does the AI rely primarily on the Pareto criterion, the Kaldor-Hicks criterion, majority voting, or some other criterion?
   d. Would changing the distribution of benefits while keeping total benefits unchanged affect its recommendation? Test this by changing the benefits across residents.
   e. Is the final decision about whether the project should be undertaken purely an economic calculation? Explain which parts of the decision require normative judgment.

Resident	Individual Benefit	Cost Share
A	800	500
B	800	500
C	300	500
D	350	500
E	450	500
Total	**2700**	**2500**